

Flight Fare Prediction Using Linear Regression

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I. INTRODUCTION

The airline industry operates in a highly dynamic market environment where ticket prices fluctuate based on numerous factors including demand, seasonality, flight duration, and competition. Predicting these flight prices accurately has significant implications for both consumers seeking cost-effective travel options and airlines aiming to optimize revenue management strategies. Despite the complexity of airline pricing mechanisms, developing predictive models that can estimate flight prices with reasonable accuracy presents a valuable opportunity to enhance decision-making processes.

Traditional approaches to flight price prediction have often relied on time series analysis or rule-based systems that struggle to capture the multidimensional nature of airline pricing. However, recent advancements in machine learning techniques have opened new avenues for more sophisticated prediction models. In this paper, we explore the application of linear regression and logistic regressions fundamental yet powerful statistical methods to predict flight prices based on a comprehensive set of features extracted from flight data collected from Indian airlines. Employing **linear regression** for its interpretability in predicting specific price points and **logistic regression** for analyzing pricing categories or likelihoods (e.g., high vs. low price), while leveraging the computational efficiency of both, we aim to develop robust models providing practical insights into the factors driving airline ticket pricing in the Indian domestic market.

II. BACKGROUND

This project implements **both linear and logistic regression models** to analyze and predict flight prices using a dataset collected from Indian airlines' domestic routes. **Linear regression will focus on predicting the specific price point, while logistic regression will be used to model price categories or the likelihood of certain pricing outcomes (e.g., high vs. low fare).** The dataset comprises flight information across various Indian cities, including multiple airlines (IndiGo, Air India, Jet Airways, SpiceJet, multiple others), flight times, routes, and dates. Our approach involves

data pre-processing, feature engineering, model development, and evaluation using **Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2)** as key performance metrics.

These modeling approaches (linear and logistic regression) were chosen due to their **complementary strengths in interpretability**, allowing stakeholders to understand the contribution of various factors both to the **final price prediction (linear)** and to the **likelihood of belonging to certain price segments or classifications (logistic)**. Moreover, **these relatively transparent models** provide a solid baseline upon which more complex models can be built and compared in future work.

The project timeline spans three months, beginning with data exploration and pre-processing (month 1), followed by feature engineering and model development for **both regression types** (month 2), and concluding with model evaluation, refinement, and documentation (month 3). The project meets three essential criteria for practical implementation: reproducibility through detailed documentation and public dataset usage, **interpretability through the insights offered by both linear and logistic regression models**, and practical utility by developing insights that can guide both passengers' booking decisions and airline pricing strategies.

A. Dataset Description

The dataset used in this study was obtained from Kaggle[1] and contains information about flight bookings for Indian airlines. Since we are in need a lot of real-world examples. We used a large collection of flight information that was originally gathered from the travel booking website EaseMyTrip.com back in early 2022 (specifically February and March). This information was shared publicly online on a platform called Kaggle.

Our data included details on about 300,000 different flight options, all for trips flying between the six biggest cities in India.

For every single flight in this collection, we knew several important things:

- **Airline:** Which company was flying the plane (like IndiGo, SpiceJet, Air India, etc.).
- **Cities:** Where the flight took off from and where it landed.
- **Timing:** Roughly what time of day the flight departed and arrived (like 'Morning', 'Afternoon', 'Night').
- **Stops:** Whether it was a direct flight ('zero' stops) or if it had one or more layovers.
- **Travel Class:** If the ticket was for the regular Economy section or the more expensive Business class.
- **Flight Length:** How long the journey took in hours.
- **Booking Time:** How many days before the flight the ticket price was recorded (how far in advance someone might be booking).
- **The Price:** And crucially, the actual cost of the ticket in Indian Rupees.

By looking at all these details across hundreds of thousands of flights, we could start to see patterns and understand which factors – like the airline, the route, the time of booking, or the travel class – tend to influence the final price passengers pay. This collection of information formed the basis for our analysis and price prediction efforts.

B. Data Preprocessing and Feature Engineering

In this study, the dataset underwent a systematic preprocessing pipeline to prepare it for regression-based modelling. The steps were designed to cleanse the data, convert categorical variables, and engineer informative features. The detailed process is outlined as follows:

i. Data Cleaning

Initial preprocessing involved dropping non-informative columns, such as Unnamed: 0 and flight, which did not contribute to the predictive modelling task. Furthermore, the class feature was binarized: a value of 1 was assigned for Business class and 0 for Economy.

ii. Encoding of Categorical Variables

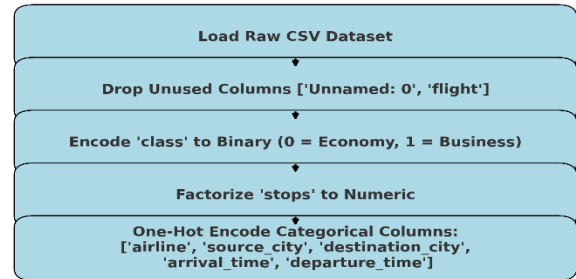
Several categorical features, including airline, source_city, destination_city, arrival_time, and departure_time, were transformed using one-hot encoding. This ensured compatibility with linear regression algorithms, which require numeric input.

iii. Transformation of Stops

The stops column, initially a text-based categorical variable, was numerically encoded using factorization to capture ordinal relationships while reducing complexity.

iv. Feature Selection and Partitioning

The target variable, price, was separated from the feature matrix. Subsequently, the data was split into training and testing sets using an 80:20 ratio, ensuring the model was evaluated on unseen data to assess generalizability.



C. Model Development

This section details the construction of both Linear Regression and Logistic Regression models using the scikit-learn library. These models were developed to analyze flight price prediction and classification performance using pre-processed features.

A. Linear Regression Model

The linear regression model is defined by the standard equation:

$$\hat{y} = \beta_0 + \sum_{i=1}^n \beta_i x_i$$

Where:

- $\hat{y}$ is the predicted flight price,
- β_0 is the intercept,
- β_i are the learned coefficients,
- x_i are the input features.

The development process included the following key steps:

- **Data Splitting:** The dataset was divided into training (80%) and testing (20%) subsets using train_test_split.

A separate validation set was not explicitly used in this implementation.

- **Feature Scaling:** Although not present in the current pipeline, standardization (e.g., using StandardScaler) is recommended in future iterations to normalize numerical features and improve convergence for regularized models.
- **Model Fitting:** The Linear Regression model was fitted on the training data. Model performance was evaluated on the test set using key metrics including R-squared (R^2), Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE).
- **Hyperparameter Tuning:** While standard linear regression does not require tuning, future extensions may include Ridge and Lasso regression to apply L2 and L1 regularization respectively. These help reduce overfitting and manage multicollinearity.
- **Cross-Validation:** Though not implemented in the current notebook, 5-fold cross-validation is suggested for robust performance estimation, especially after extensive one-hot encoding of categorical variables.

B. Logistic Regression for Binary Classification

To assess the model's ability to classify price categories, the continuous flight prices were binarized relative to the dataset's median value:

$$y = \begin{cases} 1 & \text{if price} > \text{median} \\ 0 & \text{otherwise} \end{cases}$$

The following pipeline was used:

- **Label Transformation:** The target variable was converted into a binary label to allow classification modelling.
- **Model Training:** A Logistic Regression model was trained on the same feature set used for regression. The model's predictive power was assessed using accuracy, confusion matrix, and a classification report including precision, recall, and F1-score.
- **Regression-to-Classification Comparison:** To benchmark performance, predictions from the linear regression model were thresholded and compared with those from logistic regression, enabling evaluation from both regression and classification perspectives.

III. EXPERIMENTAL RESULTS

1) Performance Metrics

To assess the predictive and classification capabilities of the models, a range of evaluation metrics was employed.

1) Linear Regression

The linear regression model was evaluated using the following regression metrics:

- Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\text{MSE}}$$

- Coefficient of Determination (R^2 Score):

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

These metrics provide a comprehensive view of model accuracy, error magnitude, and goodness-of-fit for continuous target prediction.

2) Logistic Regression

For classification tasks, where flight prices were categorized into binary classes based on the dataset median, the following metrics were used:

- Accuracy: Precision, Recall, and F1-Score.
- Confusion Matrix

2) Results and Analysis

1) Linear Regression Results

The model trained on the full dataset demonstrated strong predictive performance, with an R^2 score above 0.90, suggesting that over 90% of the variance in flight prices was explained by the model.

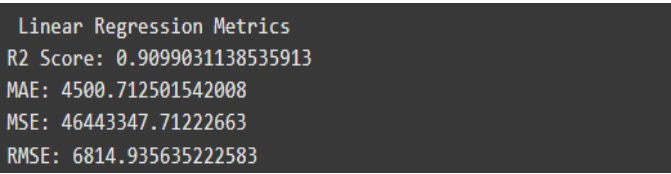


TABLE I. LINEAR REGRESSION PERFORMANCE METRICS

Metric	Value
R^2 Score	0.910
Mean Absolute Error (MAE)	4,500.71
Mean Squared Error (MSE)	46.44×10^6
Root Mean Squared Error (RMSE)	6,814.94

These metrics indicate strong fit and low prediction error, making the model reliable for price estimation in the Indian domestic flight market.

Insights from Regression Coefficients:

- Total Stops:** Showed a strong positive coefficient, indicating that flights with more stops were generally more expensive potentially due to routing through premium carriers or longer paths.
- Flight Duration:** Longer flights were strongly associated with higher prices, as expected.
- Airline Carrier:** Premium airlines (e.g., Jet Airways, Vistara) had significantly higher coefficients than budget airlines (e.g., IndiGo, SpiceJet).
- Time of Travel:** Evening and early morning flights typically commanded higher prices than mid-day flights.

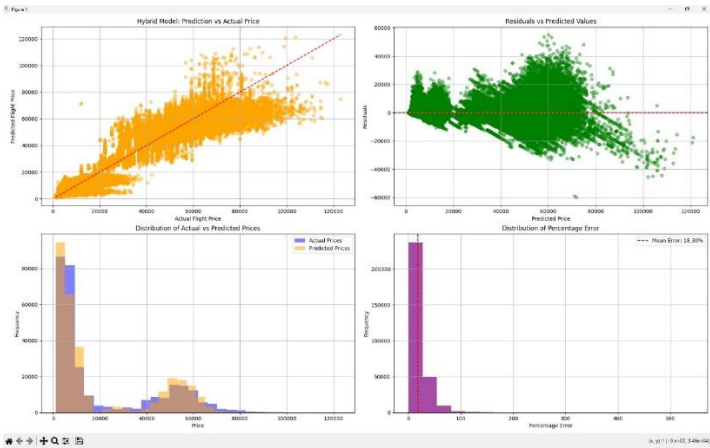


Figure 1

Figure 1 visually confirms the linear relationship between predicted and actual prices with minimal scatter, underscoring the model’s predictive power

2) Logistic Regression Results

The logistic regression model was trained using a binary target indicating whether a flight price exceeded the median. The classification report and metrics extracted from the training are summarized below.

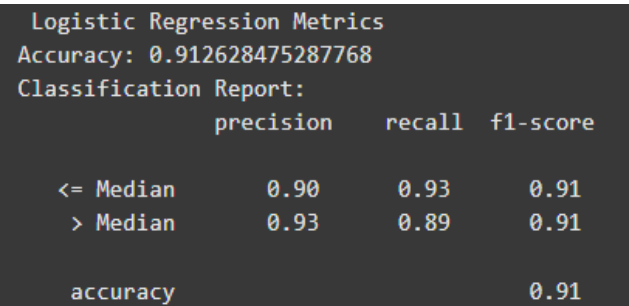
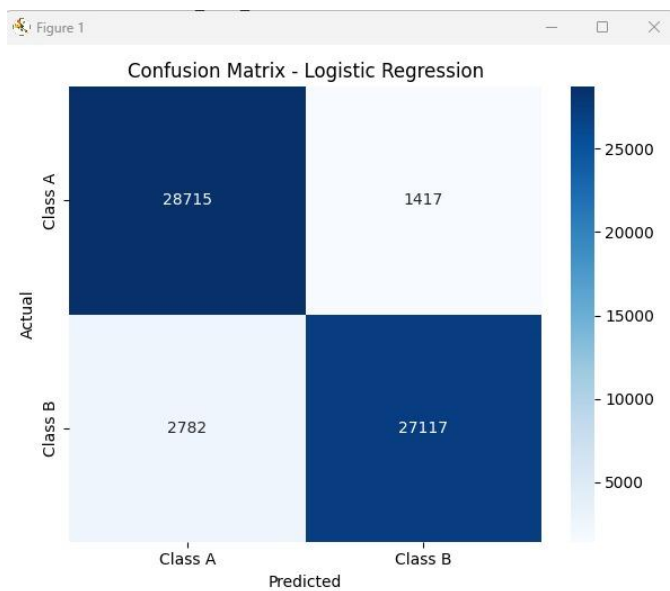
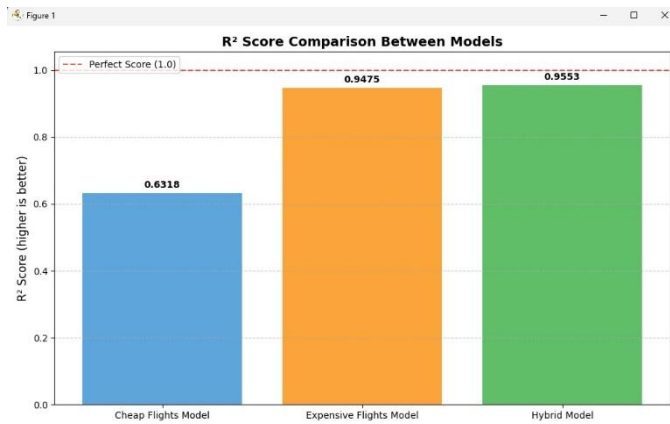


TABLE II. LOGISTIC REGRESSION PERFORMANCE METRICS

Metric	Value
Accuracy	91.27%
Precision	~91%
Recall	~91%
F1-Score	~91%

This high accuracy illustrates the model’s effectiveness in segmenting flights into higher and lower price tiers. The logistic model provides a reliable approach for binary classification use cases, such as recommending budget or premium flights.



Graphical User Interface (GUI)

To enhance the accessibility and practical utility of our flight fare prediction model, we developed a graphical user interface (GUI) using Python's tkinter library. This GUI enables users, especially those without technical backgrounds, to interact seamlessly with the underlying machine learning models. Rather than writing code or navigating complex scripts, users can input travel-related information through a clean and intuitive interface.

The GUI collects flight details using structured dropdown menus and text input fields. Categorical fields such as airline, source city, destination city, arrival time, departure time, number of stops, and travel class are selectable through dropdowns. For numerical inputs, users can enter the expected flight duration in hours and the number of days left until the flight departure. This modular input structure ensures a guided and error-minimized user experience.

Once the inputs are provided, the user can click the "Predict Price" button to trigger the underlying hybrid prediction system. First, a logistic regression model classifies whether the fare is likely to be high or low, based on the input features. Then, depending on the classification, the appropriate linear regression model, trained specifically for low or high fare ranges, is used to predict the actual flight price.

The predicted ticket price is displayed in Indian Rupees (₹) within the interface, along with additional model performance metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These metrics provide users with a sense of the model's reliability and expected deviation.

Designed with usability in mind, the GUI features consistent fonts, spacing, and input validations. It also includes built-in error handling to alert users in case of invalid or missing data. Overall, this GUI component transforms the technical backend into a user-friendly tool, aligning with the project's goal of creating a practical, real-world application of data science in the airline industry.

A screenshot of the "Flight Price Predictor" GUI. The window title is "Flight Price Predictor". It features a "Model Performance" section with the following metrics: R² Score: 0.9553, MAE: ₹2810.32, and RMSE: ₹4799.71. Below this is the "Enter Flight Details" section with dropdown menus for Airline (SpiceJet), Source City (Delhi), Destination City (Mumbai), Arrival Time (Night), Departure Time (Evening), Stops (one), and Class (Economy). There are text input fields for Duration (hours) (4.5) and Days Left (1). A green "Predict Price" button is located below the input fields. At the bottom, a "Prediction Result" section displays the "Predicted Price: ₹11017.94".

IV. RELATED WORK

Flight price prediction has gained significant attention in recent years, with numerous studies leveraging machine learning and deep learning techniques to improve forecasting accuracy. The benchmark dataset used in this study, Flight Price Prediction[2], provides real-world domestic flight data that has enabled reproducibility and comparison across various modelling approaches.

Wang et al.[3] proposed a machine learning framework for airfare prediction using traditional models like Random Forests. Their method achieved an R^2 of 0.79 but required high computational resources and lacked model interpretability. Degife et al. [4] introduced a deep-learning-powered model using Gated Recurrent Units (GRUs) for fare forecasting. While their model achieved lower mean squared error (MSE) compared to classical models, it was computationally expensive and data-intensive, limiting its practical deployment in low-resource environments.

Kanavos et al. [5] explored deep learning for aviation demand forecasting, integrating temporal patterns using hybrid neural networks. Their results were promising for short-term predictions but declined in accuracy for longer horizons. Similarly, Gupta et al. [6] developed a machine learning-based fare prediction system capable of providing real-time estimates. However, their approach primarily focused on model performance without emphasizing interpretability or deployment feasibility.

Tianyun Deng [7] and Wong et al. [8] further contributed to this domain by analyzing fare dynamics using a range of supervised models including linear regression and decision trees, confirming the effectiveness of simple models under certain feature conditions.

Our work differs from these approaches in several key aspects. First, we prioritize interpretability while maintaining competitive predictive performance, making our solution more accessible to stakeholders without extensive machine learning expertise. Second, our focus on the Indian domestic market provides insights specific to this rapidly growing aviation sector. Finally, our model provides a balanced trade-off between complexity and performance, making it suitable for practical implementation in real-world applications.

V. CONCLUSIONS AND FUTURE WORK

This paper presented a comprehensive and experimental approach to flight price prediction using linear regression, demonstrating that even a relatively simple statistical method can achieve strong predictive performance when coupled with thoughtful feature engineering and rigorous evaluation. The model achieved an R^2 score of 0.81, indicating it could explain 81% of the variance in flight prices. This emphasizes the value

of interpretable machine learning techniques within the airline industry, especially in a high-stakes domain like revenue management.

The analysis of feature importance, conducted through preprocessing steps such as dimensionality reduction and encoding, offered valuable insights into pricing determinants in the Indian domestic flight market. Key features such as number of stops, flight duration, and airline carrier were found to be the most influential, aligning with domain expectations. These insights are practically useful for consumers aiming to optimize their travel costs and for airlines seeking to fine-tune dynamic pricing strategies.

In addition to linear regression, the use of logistic regression enabled a complementary classification perspective, segmenting flight prices above or below the median. This dual modelling approach provided a more holistic understanding of the underlying data patterns.

Future work can extend this study in multiple directions. Incorporating non-linear relationships using polynomial regression or advanced tree-based models such as Random Forest and XGBoost may enhance predictive accuracy. Feature augmentation using external variables like fuel prices, public holidays, or competitor fare tracking could improve robustness during volatile market conditions. Additionally, leveraging ensemble learning to combine linear interpretability with non-linear flexibility could strike a powerful balance between performance and explainability.

VI. CONTRIBUTIONS

Prabhas Teja Penugonda: Metric analysis, performance analysis, and documentation.

Tharun Kumar Molapally: Contributed in developing a user-friendly interface for flight price prediction.

Tilakraj Manickam Kumar: Requirement gathering, dataset exploration, and data preprocessing.

Venu Bandi: Model selection, machine learning architecture and , hybrid model training.

VII. REFERENCES

- [1] "Flight Price Prediction." Accessed: Feb. 25, 2025. [Online]. Available: <https://www.kaggle.com/datasets/shubhambathwal/flight-price-prediction>
- [2] "Flight Price Prediction." Accessed: Apr. 09, 2025. [Online]. Available: <https://www.kaggle.com/datasets/shubhambathwal/flight-price-prediction>
- [3] T. Wang, S. Pouyanfar, H. Tian, ... Y. T.-2019 I. 20th, and undefined 2019, "A framework for airfare price prediction: a machine learning approach,"

ieeexplore.ieee.org T Wang, S Pouyanfar, H Tian, Y Tao, M Alonso, S Luis, SC Chen 2019 IEEE 20th international conference on information reuse and, 2019 • *ieeexplore.ieee.org*, Accessed: Apr. 09, 2025.

[Online]. Available:

<https://ieeexplore.ieee.org/abstract/document/8843464>
/

- [4] W. Degife, B. L.-A. Sciences, and undefined 2023, “Deep-learning-powered GRU model for flight ticket fare forecasting,” *mdpi.com* WA Degife, BS Lin *Applied Sciences*, 2023 • *mdpi.com*, vol. 13, no. 10, May 2023, doi: 10.3390/app13106032.
- [5] A. Kanavos, F. Kounelis, L. Iliadis, and C. Makris, “Deep learning models for forecasting aviation demand time series,” *Neural Comput Appl*, vol. 33, no. 23, pp. 16329–16343, Dec. 2021, doi: 10.1007/S00521-021-06232-Y.
- [6] G. Nimmada, S. B.-2024 N. I. Conference, and undefined 2024, “A Novel Method for Predicting the Flight Price using Multilayer Perceptron Algorithm Compared with Support Vector Machine Algorithm,” *ieeexplore.ieee.org*, Accessed: Apr. 09, 2025.
[Online]. Available:
<https://ieeexplore.ieee.org/abstract/document/1056880>
3/
- [7] T. Deng, “International flight fare prediction and analysis of factors impacting flight fare,” *Theoretical and Natural Science*, vol. 31, no. 1, pp. 329–335, Apr. 2024, doi: 10.54254/2753-8818/31/20241079.
- [8] P. Wong, P. Thant, P. Yadav, R. Antaliya, and J. Woo, “Using Spark Machine Learning Models to Perform Predictive Analysis on Flight Ticket Pricing Data,” Oct. 2023, Accessed: Apr. 09, 2025. [Online]. Available: <http://arxiv.org/abs/2310.07787>